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CLINICAL FEATURE
ORIGINAL RESEARCH

Previous foot injuries associated with a greater likelihood of Achilles tendon ruptures in professional American football players

Michael K. Krill[a,b,c], Joshua Hoffman[a], JaeWon Yang[d], Johnathan D. Hodax[e], Brett D. Owens[e] and Timothy E. Hewett[f,g,h,i]

[a]James Crane Sports Medicine Institute, Motion Analysis and Performance Laboratory, The Ohio State University Wexner Medical Center, Columbus, OH, USA; [b]Charles E. Schmidt College of Medicine, Florida Atlantic University, Boca Raton, FL, USA; [c]Physical Medicine and Rehabilitation, Department of Neurology, Division of Neurorehabilitation, Washington University in St. Louis, St. Louis, MO, USA; [d]Warren Alpert Medical School of Brown University, Providence, RI; [e]Department of Orthopaedic Surgery, Warren Alpert Medical School of Brown University, Providence, RI, USA; [f]Sports Medicine Center, Mayo Clinic, Rochester, MN, USA; [g]Department of Orthopedic Surgery, Mayo Clinic, Rochester, MN, USA; [h]Department of Physiology and Biomedical Engineering, Mayo Clinic, Rochester, MN, USA; [i]Department of Physical Medicine & Rehabilitation, Mayo Clinic, Rochester, MN, USA

ABSTRACT

Objectives: The purpose of this study was to build on current understanding of Achilles tendon (AT) ruptures in football through the examination of lower extremity injury rates (IR) in the season prior to AT rupture.

Methods: An online review was conducted to document AT ruptures in the National Football League (NFL) from the 2010–11 through 2016–17 seasons. Player-specific information was gathered through the online search, injury reports, or video analysis. Injury incidence was recorded for each season. Descriptive statistics were calculated for all variables. IRs per 1000 athlete-exposures with 95% confidence intervals (CI) were established. Injury rate ratios (IRR) with 95% CI were calculated to evaluate differences in IR with statistical significance established at $p < 0.05$.

Results: From the 2010–11 to 2016–17 NFL seasons, 109 AT ruptures were identified. 36 athletes participated in the NFL the season prior and sustained a total of 40 injuries. Thirty-two (32/40, 80%) of the injuries involved the lower extremity. The most frequent body parts injured in the NFL season prior to AT rupture were the knee (n = 8, 22.5%), upper leg and thigh (n = 7, 17.5%), lower leg and shin (n = 5, 12.5%), and ankle (n = 5, 12.5%). The overall IR for the AT rupture group was not significantly different than the NFL group (IRR: 0.94, 0.68–1.31 95% CI, p = 0.73). The AT rupture group demonstrated a significantly increased rate of foot injuries compared to the NFL cohort (IRR: 2.58, 1.16–5.77 95% CI, p = 0.02).

Conclusion: There was an increased identified incidence of AT ruptures in the NFL from 2010 to 2017 compared to AT ruptures reported from 1997 to 2002. There was no significant difference in lower extremity IR the season prior to AT rupture. Only foot injuries demonstrated an increased IR the season prior to AT rupture compared to an NFL cohort. Eleven athletes sustained 2 AT ruptures and all 11 players sustained the subsequent AT rupture to the contralateral limb.

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Introduction

Achilles tendon (AT) ruptures are an unfortunate growing trend in football [1–4]. AT ruptures are common during eccentric loading of the triceps surae prior to explosive acceleration or change of direction [5,6]. A 2009 review by Parekh et al. reported an overall incidence of 31 AT ruptures from 1997 to 2002 in the National Football League (NFL) [5]. Despite increased attention regarding NFL injuries, recent epidemiologic studies of AT ruptures in the NFL are limited [2,4,7–13]. AT ruptures significantly affect player performance [5,14–16]. For example, Parekh et al. created a power formula to compare statistical performance pre- and post-AT rupture of 31 NFL athletes and discovered that players demonstrated a decrease of >50% in statistical performance after injury [5]. Mai et al. calculated the return-to-play (RTP) rate of NFL players after major orthopedic procedures (i.e. ACL rupture, knee microfracture surgery, ankle fracture open reduction internal fixation, lumbar discectomy) and found the RTP rate after AT ruptures to be only 72.5%, the second lowest RTP rate [17]. AT ruptures pose a significant risk to the performance and longevity of elite football players [5,17].

The potential significant long-term effects of AT ruptures pose a unique challenge [18–21]. In addition to the low RTP rate, the average recovery time before RTP in the NFL after an AT rupture is 375 days, significantly longer than the mean recovery in the NFL of most common orthopedic procedures (i.e. discectomy, spinal fusion, ligament reconstruction, intramedullary nailing, open reduction and internal fixation) [17]. Currently, the average career length in the NFL is only

CONTACT Timothy E. Hewett hewett.timothy@mayo.edu Department of Orthopedics, Mayo Clinic, 200 First Street SW, Rochester, MN 55905
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3.3 years and declining, which makes any extended absence a significant risk to a career [13,22]. The effects of an AT injury continue even after RTP [17]. NFL athletes participate in significantly fewer games in their first season after AT rupture [17]. Fortunately, there is no difference in game participation rates in subsequent seasons [17]. These lead to long-term financial and competitive disadvantages [5,17]. Overall, NFL careers are limited to an average of only 27.3 games after an AT rupture, roughly 1.6 seasons [17]. Elite athletes experience the worst rate of return to pre-AT rupture activity levels on patient-reported functional outcome questionnaires [17]. With an extended recovery period and shortened NFL career, considerable financial ramifications negatively impact NFL athletes.

A thorough understanding of the risk factors for AT ruptures is paramount to reduce AT rupture rates and protect NFL athletes from the potential negative effects from the post-rupture recovery period. After establishment of a thorough understanding of AT ruptures in professional football, next steps will be to create screening protocols to identify high-risk athletes and establish training regimens to reduce risk of rupture. The purpose of this study was to build on current understanding of AT ruptures in football through the examination of lower extremity IR the season prior to AT rupture. The hypothesis tested was that players who suffered an AT rupture would demonstrate a greater incidence, proportion and IR of lower extremity game injuries the season prior to AT rupture compared to the rest of the NFL.

Materials and methods

An online review was conducted to document AT ruptures in the NFL from the 2010–11 through 2016–17 seasons. Online searches were completed through Google.com with the combination of the words 'Achilles', 'tendon', player names, teams, and year of injury. Additional NFL, team, and player information was retrieved from Rotoworld.com, NFL.com, ESPN.com, SBNation.com, SI.com, and reports from local sports reports and blogs. Weekly published team injury reports and injury reserve lists were gathered to identify athlete participation, injury status, and body part injured through published injury reports. Injury reports that reported a player did not participate due to a coach's decision or undisclosed reasons resulted in further internet searches to identify the reason for missed participation. This was an effort to identify any player who had an injury that was not listed on the team injury reports. In addition, updates were made to the injury reports if interviews and news reports differed. All injuries listed to a body part inferior to the knee were evaluated through an online search to identify all AT ruptures. Only injuries with corresponding injury reports, news articles, or interviews that confirmed AT rupture were included.

Player-specific information was gathered through the online search, injury reports, or video analysis and included team, position, approximate date of the injury, body part injured, and games missed. Injury incidence was recorded for each season. A season was defined as the day after the previous Super Bowl until the day of the Super Bowl of each season. NFL athletes participate in practices each week of the season, but these NFL practices have a reported very low IR [23,24]. For this reason, an injury was defined as any injury documented on the injury report that resulted in the athlete not participating for a minimum of one game. An athlete-exposure (AE) was defined as one athlete participating in one game.

An entire analysis of all NFL team injury reports from 2010 to 2017 were included to calculate injury rates (IRs) across the entire NFL. All injury documentation, methods, and definitions were maintained for the NFL control group. For each NFL game, one team has 46 active players dress who are allowed to participate. For our study analysis, 32 NFL teams participate in 15 games and 1 bye week with mandatory published injury reports for the following week to document injuries. Per our methodology, this resulted in 22,080 AE each season for the NFL group.

Descriptive statistics were calculated for all variables. IR per 1000 AE with 95% confidence intervals (CI) were calculated. Injury rate ratios (IRR) with 95% CI were calculated to evaluate differences in IR with statistical significance established at $p < 0.05$.

Results

From the 2010–11 to 2016–17 NFL seasons, 109 AT ruptures were identified (Table 1). Of the 109 AT ruptures, 21 athletes did not play in the NFL the previous season (i.e. free agent, National Collegiate Athletic Association student-athlete, Canadian football league player, etc.). The mean age at the time of injury for the AT ruptures was 26 ± 3 years. That left 88 athletes who played in the NFL the season prior to their AT rupture. Of the remaining 88 AT ruptures, 52 sustained no documented injuries the NFL season prior to AT rupture and 36 athletes experienced at least one missed game due to injury. In these 36 monitored athlete-seasons prior to AT rupture, a total of 40 injuries were documented (Table 1).

Table 1. Breakdown of injuries documented in the season prior to Achilles tendon rupture.

Body part injured	Incidence (proportion)
Head/neck	**2 (5%)**
Head	*1*
Neck	*1*
Shoulder/chest	**2 (5%)**
Shoulder	*1*
Pectoral	*1*
Wrist/hand	**2 (5%)**
Wrist	*1*
Thumb	*1*
Back	**2 (5%)**
Upper leg/thigh	**7 (17.5%)**
Hamstring	*5*
Groin	*2*
Knee	**9 (22.5%)**
Lower leg/shin	**5 (12.5%)**
Calf	*2*
Achilles	*3*
Ankle	**5 (12.5%)**
Foot	**6 (15%)**
Foot	*4*
Toe	*2*
Head, spine, and Back	**4 (10%)**
Upper extremity	**4 (10%)**
Lower extremity	**32 (80%)**
Total number of injuries	**40**

Thirty-two (32/40, 80%) of the injuries involved the lower extremity and only eight (8/40, 20%) injuries to the head, back, or upper extremity. The most frequent body parts injured included the knee (n = 9, 22.5%), upper leg and thigh (n = 7, 17.5%), lower leg and shin (n = 5, 12.5%), and ankle (n = 5, 12.5%).

These previous injuries from the AT rupture group were broken down further into when an injury was sustained each year. Five players suffered season-ending injuries in the off-season or preseason and missed the entire regular season the year before AT rupture (Figure 1). Of the five season-ending injuries, three (n = 3, 60%) were to the AT, one (n = 1, 20%) to the knee, and one (n = 1, 20%) to the calf (not AT related per news documentation and interviews available). One player sustained an undisclosed injury the year prior to AT rupture. We attempted to determine the cause and were unable. This AT rupture was excluded from the previous season injury history analysis. This left 31 players who competed in the NFL and made an active team roster for at least one game the season before AT rupture. These 31 players sustained 35 total injuries during the season from 301 AE (Mean: 9.71 AE) as shown in Table 2. The 52 players who did not sustain an injury participated in 656 AE (Mean: 12.62 AE). The IR for all athletes that sustained an AT rupture was 36.57 per 1000 AE (25.87–50.30 95% CI).

There were 11 players that suffered two AT ruptures during the review. Of the 11 players with multiple AT ruptures, three (n = 3) players sustained an AT rupture in consecutive seasons. In all three cases, the AT rupture occurred to the contralateral AT. Overall, all 11 players with multiple AT ruptures sustained the second AT rupture to the contralateral AT. There were no ipsilateral AT ruptures.

Overall, the NFL comparison group sustained 4228 injuries during the 2010–11 to 2016–17 seasons. The majority of documented NFL injuries were to the lower extremity (n = 2,887, 68.3%). Of the lower extremity injuries, 216 were to the lower leg and shin (5.1%). The overall IR for the AT rupture group was not significantly different than the NFL group (IRR: 0.96, 0.40–2.30 95% CI, p = 0.79). The AT rupture group demonstrated a significantly increased rate of foot injuries compared to the NFL cohort (IRR: 2.58, 1.16–5.77 95% CI, p = 0.02). Otherwise, there was no difference between the IRs sustained to the lower extremity (all $p > 0.05$). There was also no significant difference in IRs for all injuries combined at or below the knee (p = 0.32).

Discussion

This study evaluated previous lower extremity injury in the prior NFL season as a risk factor for AT ruptures in professional football players. This study found that 40.9% of NFL players who suffered an AT rupture participated in the NFL the year prior and sustained an injury that held them from game competition, primarily to the lower extremity (80%).

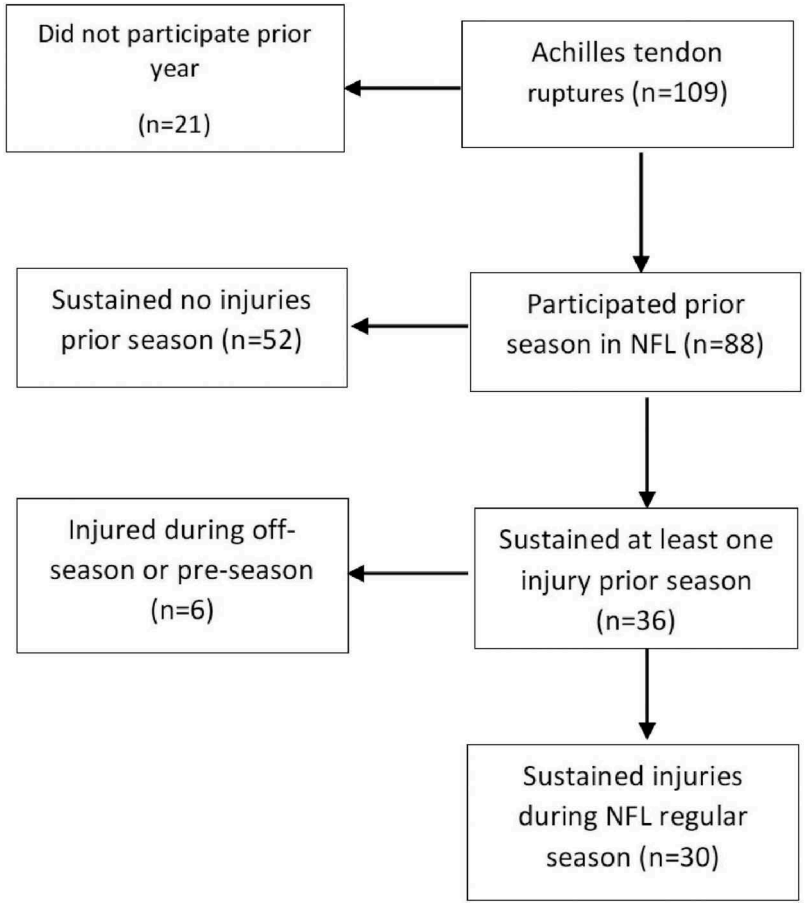

Figure 1. Player participation the season prior to Achilles tendon rupture.

Table 2. Prior lower extremity injury comparison of an NFL cohort and Achilles tendon rupture group.

	AT rupture group	NFL cohort	IRR	*p*-Value
Overall Injuries	35	6002	0.94 (0.68–1.31)	0.73
IR (95% CI)	36.57 (25.87–50.30)	38.83 (37.86–39.82)		
LE injuries	28	4107	1.10 (0.76–1.58)	0.62
IR (95% CI)	29.26 (19.82–41.72)	26.57 (25.77–27.39)		
Knee injuries	8	1190	1.09 (0.54–2.17)	0.82
IR (95% CI)	8.36 (3.88–15.87)	7.70 (7.27–8.15)		
Lower leg and shin injuries	2	310	1.04 (0.26–4.18)	0.95
IR (95% CI)	2.09 (0.35–6.91)	2.01 (1.79–2.24)		
Ankle injuries	5	845	0.96 (0.40–2.30)	0.92
IR (95% CI)	5.22 (1.91–11.58)	5.47 (5.11–5.85)		
Foot injuries	6	374	2.58 (1.16–5.77)	0.02
IR (95% CI)	6.27 (2.54–13.04)	2.42 (2.18–2.68)		
Knee, lower leg, and shin, ankle, and foot injuries	21	2719	1.24 (0.81–1.90)	0.32
IR (95% CI)	21.94 (13.95–32.97)	17.59 (16.94–18.26)		

The AT rupture group had an overall IR of 36.57 per 1000 AE the season before AT rupture, slightly lower than the NFL IR (38.83 per 1000 AE); however, the AT rupture group demonstrated a slightly increased IR to the lower extremity. There was no difference in IR to the lower leg and shin the season prior to AT rupture, although this may have been confounded as three injuries to the AT and one injury to the calf occurred in the off-season or preseason and would not appear in this analysis of game injuries. None of those differences were significant and there was no difference in IRs to the lower leg and shin, upper leg and thigh, knee, or ankle. Injuries to these LE body regions are not significant alone to be a risk factor to identify players predisposed for AT ruptures. The AT rupture group demonstrated a significantly increased IR with foot injuries compared to the NFL group. This may be an incidental finding due to the vague nature of injury reports that do not require extensive injury details or low frequency of foot injuries. Although, this may be associated with the relationship between foot biomechanics and Achilles tendinopathy that may predispose the tendon to rupture.

Several studies have reported potential risk factors that may lead to an increased rate of Achilles tendinopathy (pathologic change in the tendon), not specifically AT ruptures [25–33]. There can be multiple causes of tendinopathy [31,33]. Tendinosis is the degenerative condition that lacks clinical or histologic signs of inflammation change; whereas, tendinitis involves a histopathologic confirmation of inflammatory processes although can be used differently in a clinical context [31,33,34]. One major predisposing factor is lower extremity biomechanical or training errors [25–27]. Excessive pronation may lead to repetitive microtrauma of the AT through two proposed mechanisms [25–27]. The first mechanism is a whipping action of the AT with excessive eversion motion of the hindfoot [25,26]. Excessive hindfoot motion leads to increased forces on the medial aspect of the AT that can lead to microtears with repetition [25]. The second proposed mechanism is asynchronous movement between the foot and ankle during the stance phase due to abnormal pronation that may lead to a 'wringing' effect within the AT [25,26]. This 'wringing' effect impinges the vascular system leading to vascular impairment within the tendon [25,26]. In addition, there may be an increase in tensile strength on the tendon in the transverse plane due to the lack of control of the distal tibia and decreased muscular function [25,26,28]. Over time these proposed repetitive mechanisms lead to degenerative changes of the AT [25,26], where an ultimate stress can lead to acute tendon failure [29]. The most common location for acute AT ruptures is roughly 2–6 cm proximal to the insertion on the calcaneus [29]. Abnormal tendon changes have been observed in elite Danish soccer players [35]. In a 2002 study, Fredberg et al. performed preseason ultrasound (US) on asymptomatic AT to identify the frequency of abnormal tendon changes in asymptomatic tissue and found that 11% of asymptomatic soccer players had abnormal preseason AT US findings and experienced a 45% risk of development of symptoms of Achilles tendinosis during the season [35]. Although it is a challenge to obtain accurate injury details with regard to the foot injuries sustained by NFL players, an interesting analysis regarding the frequency of injuries involving specifically the hind- and mid-foot may provide further insight in to a significant underlying risk factor component of AT ruptures in NFL athletes.

Three of the players that suffered AT ruptures sustained an AT rupture in two consecutive seasons. In all three cases, there were no re-ruptures and all were contralateral. Over the course of the study, there were a total of 11 athletes identified with 2 AT ruptures. All 11 cases involved bilateral AT ruptures. There were no AT re-ruptures identified. Aroen et al. calculated that individuals who sustained an AT rupture demonstrated a vastly increased risk, nearly 200 times, of future contralateral AT rupture compared to the AT rupture rate in the general population [36]. Further analysis demonstrated degenerative changes to the ruptured AT soft tissue [36,37]. Although the previously described methods provide mechanisms for degenerative tissue changes to the AT, an additional consideration in the three contralateral ruptures in consecutive years is disuse atrophy from extended absence from high-level training [36]. There may be other factors as well that contribute to AT ruptures and

studies have evaluated genetic risk factors to explain multiple AT ruptures.

As reported by Mai et al., the RTP rate after AT ruptures for NFL athletes is extremely low (72.5%) compared to other orthopedic procedures in the NFL [17]. In addition, the athletes that do return experience significant reductions in statistical performance and power production in the repaired AT [17]. One possible reason for a decreased risk of re-ruptures may be that athletes who are most likely to re-rupture their AT never return to competitive football or pristine fitness levels as only those athletes who achieve an adequate recovery are able to return to competitive sport [5,6,14,15,17]. Further evaluation of NFL athletes who sustained an AT rupture and never returned to competitive football or training may improve our understanding what leads to a successful AT reconstruction as currently AT ruptures produce one of the lowest RTP rates of orthopedic procedures. A study by Orishimo et al. identified that even after successful AT repairs, some degree of AT elongation remains regardless of technique [38]. Elongation and slack in the repaired AT is a potential contributor to decreased power production and may also reduce tension, thus prevent the elite athlete from sustaining the acute force necessary to re-rupture the AT [38]. Eleven athletes sustained bilateral AT ruptures over the 8 years of our study. Understanding the biomechanics, past medical, orthopedic, and family history may help identify factors that predispose specific athletes to AT ruptures.

This study identified no clear correlation between recent orthopedic injury history in the NFL, specifically lower extremity or lower leg and shin injuries, and the incidence of AT ruptures in professional football players. However, a *post hoc* power analysis determined the AT rupture group may not have included sufficient athletes or injuries to fully power our findings of body parts with limited injuries in the AT rupture group. A continued effort to identify predisposed individuals is crucial, as the number of AT ruptures in professional football players has increased over the last 20 years. As AT ruptures continue in the NFL, additional previous injury history of players may help improve our understanding and ability to identify high-risk players. AT ruptures clearly pose a significant risk to performance and professional longevity of NFL athletes. Most importantly, further analysis of the 11 bilateral AT ruptures may assist in identification of specific predisposing risk factors in elite athletes.

Limitations

This study utilized injury records that relied entirely upon team-reported injury data. Although this information is required by the NFL and publicly reported, there are significant reporting discrepancies by different teams and coaching staffs. Injury reports provide limited details and only list the body part injured. These reports may not be entirely accurate and do not necessarily document each body part if several injuries hamper the athlete. There are numerous cases where the injury report may indicate that the injury is undisclosed. We attempted to further search undisclosed or unclear injury reports to discover the true nature of the injury and body part injured. There were occasions where increased scrutiny discovered through athlete or coach interviews that contradicted the body part listed on the injury report. In those instances, the injury report was updated and the body part injured was corrected. This could not be completed on every injury due to different amounts of coverage regarding each injury. In addition, roster management in the NFL can impact injury records. This was clearly observed near the end of the season, as teams in the playoffs may attempt to rest players to improve their health, while other teams are still competing to make the postseason. Even teams that are no longer in playoff contention may attempt to manage the roster to give their management opportunities to further evaluate players that are on the practice squad.

To confirm AT ruptures, we attempted to confirm through publicly available game participation and injury records as a player with an AT rupture does require an extended lack of participation for surgical intervention and the appropriate rehabilitative recovery. We found this to be the most accurate method to identify and confirm AT ruptures through injury reports, media reports, and player game participation records. Despite these limitations and possible inaccuracies in the data, it is important to document and report these findings in high profile sports to monitor changes in injury patterns in specific athletic populations.

The NFL does not require teams to publish weekly injury reports for the preseason or postseason (if a team is not participating). For this reason, there is limited available information for a player who is inactive in week 1 of the regular season. This lack of participation in week 1 could be due to injuries, extended postsurgical recovery, or contract-related complications from last season, off-season activities, non-football activities, or preseason practices or games. For these reasons, the injury reports for week 1 were not included in this analysis. In addition, no injuries were included that were sustained during the last game of the regular season, since not all teams are required to complete injury reports the first week of the playoffs. Any team that did not participate in the playoffs would not have to officially document these injuries. If an injured athlete returned to the active game roster after an injury, any future missed games were considered an additional injury, even if the same body part was documented. Although it may have been the same injury that kept the athlete out in both instances, the athlete was healthy enough to return to game competition.

There may have been limited accuracy as to the timing and extent of injuries. Previous studies have identified that most injuries occur in games and a majority of in-week exposures are walk-throughs or noncontact exposures that demonstrate a low IR compared to games and live contact exposures [23,24,39–41]. The current definitions and methods focused only on game exposures due to the low intensity and documented IR of in-week practices. The current IR analysis did not produce exact AE for the entire NFL and was completed with the understanding that each team has 46 active players for each game. Due to the small active roster allowed each game, it is to each team's benefit to fully utilize the active roster each game. As injury reports are not required to be published for the preseason or after the final week of the regular season for teams not in the wildcard round, this study focused on injury reports from

regular season weeks 2 through 17 to monitor game injuries from the previous week.

Conclusion

There was an increased identified incidence of AT ruptures in the NFL from 2010 to 2017 compared to AT ruptures reported from 1997 to 2002. There was no significant difference in lower extremity IR the season prior to AT rupture. Only foot injuries had an increased IR the season prior to AT rupture compared to an NFL cohort. Eleven athletes sustained 2 AT ruptures and all 11 players sustained the subsequent AT rupture to the contralateral limb.

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Declaration of interest

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